



Quantitative Data Analysis

Session 1 — Finding Data & Univariate Analysis

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EM Normandie — Programme Grande École

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- Teaching **Quantitative Data Analysis** at EM Normandie

How I want to work with you

I will not ask you to memorise formulas.

I will ask you to **find real data**, analyse it, and **defend your conclusions** to a manager.

That is the entire course.

The course in one slide

Session	Method	Question it answers
1 <i>today</i>	Univariate descriptive analysis	What does my data look like?
2	Chi-square & Pearson correlation	Are two variables related?
3	ANOVA + post-hoc tests	Do groups differ from each other?
4	Simple & multiple regression	What drives my outcome?

14 hours face-to-face · 36 hours independent work · 5 ECTS

Software: **Jamovi** (free). One dataset — *yours* — carried through all four sessions.

What you must be able to do by session 4

1. **Identify and justify** the appropriate statistical method for a given managerial problem
2. **Implement** it in data analysis software
3. **Interpret** the statistical output accurately
4. **Draw conclusions** and formulate actionable managerial recommendations

Notice what is *not* on this list: calculating anything by hand.
The software computes. **You decide, interpret, and recommend.**

Today: 3 hours 30

Time	What happens	Mode
0:00 – 0:15	Introduction, course contract	Lecture
0:15 – 1:30	Theory: variables, descriptive statistics, distributions	Lecture + quiz
1:30 – 1:45	<i>Break</i>	
1:45 – 2:45	Data hunt: find your own dataset	Teams of 3
2:45 – 3:15	Jamovi lab: build and describe your database	Hands-on
3:15 – 3:30	Approval gate + submission of Artefact 1	Checkpoint

You do not leave this room without an approved dataset. It is the foundation of every session that follows, and of your coursework mark.



Why this matters

From data to decision



Sessions 1–4 live in the middle of this chain.

But you are graded on **both ends**: the question you ask, and the decision you recommend.

Why managers need this

Without data

- “I *think* our customers prefer. . .”
- “It *seems* sales are up”
- “Everyone *knows* that segment is loyal”

With data

- “56% of our buyers choose the regional brand”
- “Spending rises 12 € per additional visit”
- “Retirees score 15.6/20 on loyalty; students 4.8”

The second column wins budget arguments. The first does not.

That is the whole professional case for this course.

The protocol — your answer template all term

1. **IDENTIFY** — Which test? Justify from the **nature of the variables**.
2. **HYPOTHESISE** — State H_0 and H_1 in business language.
3. **DECIDE** — Is $p < 0.05$? Then: **how strong** is the effect?
4. **RECOMMEND** — What does the manager do on Monday morning?

Students lose marks at steps **1** and **4**. Almost never at step 3.
Computing is easy. **Choosing** and **concluding** are the skill.



Variables: the foundation

First: what is one row of your data?

One row = one _____

This is called the **unit of observation**. If you cannot finish that sentence, you do not understand your dataset yet.

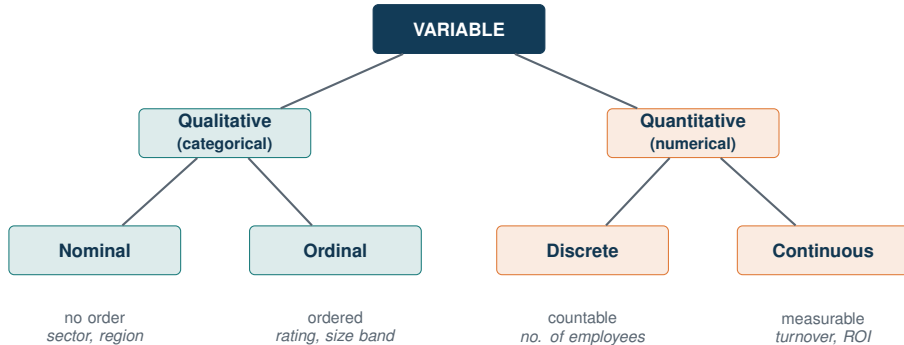
Examples

- one row = one **customer**
- one row = one **firm**
- one row = one **firm-year**
- one row = one **country**
- one row = one **transaction**

Vocabulary

- **Observation** (row) — one individual case
- **Variable** (column) — one measured characteristic
- N — number of observations

The variable taxonomy



This single diagram decides **every test** you will use in sessions 2, 3 and 4.

Nominal

Names or labels. **No natural order.**

Examples:

- Industry sector
- Country of listing
- Brand purchased
- Legal form (SA, SARL, SAS)

You may count them. You may **not** average them.

Ordinal

Ordered categories, but **unequal gaps.**

Examples:

- Credit rating (AAA, AA, A . . .)
- Satisfaction 1–5 stars
- ESG grade (A+ to D–)
- Firm size band

AA to A is not the same “distance” as B to C.

Discrete

Countable values, usually integers.

Examples:

- Number of employees
- Number of stores
- Monthly visits
- Number of board members

Continuous

Any value within a range.

Examples:

- Turnover (€)
- ROI (%)
- Share price
- CO₂ emissions per unit

Interval vs ratio. Temperature in °C is *interval*: 20° is not “twice” 10°, because zero is arbitrary. Turnover is *ratio*: zero means none, and 2M€ really is twice 1M€.
Correlation and regression need interval or ratio data.

Quiz — name the variable type

1. Customer satisfaction, 1–5 stars
2. Number of stores in a region
3. Postcode (75008, 76600)
4. Time spent on a website, in seconds
5. Moody's credit rating
6. Temperature in °C
7. Annual turnover in €
8. Loyalty score, 0–20

Take 3 minutes. Discuss with your neighbour.

Quiz — answers

Variable	Type	Why / trap
Satisfaction 1–5 stars	Ordinal	The classic trap. Gaps between stars are not equal.
Number of stores	Discrete quant.	Countable, integer.
Postcode	Nominal	Digits, but not a number. Never average a post-code.
Time on site (seconds)	Continuous quant.	Ratio scale, true zero.
Moody's rating	Ordinal	Ordered, unequal gaps.
Temperature °C	Continuous (<i>interval</i>)	Arbitrary zero: no ratios.
Annual turnover	Continuous (<i>ratio</i>)	True zero.
Loyalty score 0–20	Debatable	Ordinal in principle, treated as continuous in practice.

The last one matters: composite scores are *conventionally* treated as continuous. That is exactly why we can run an ANOVA on a loyalty score in session 3.

Why the typology decides everything

Variable X	Variable Y	Test	Session
Qualitative	Qualitative	Chi-square (χ^2) + Cramér's V	2
Quantitative	Quantitative	Pearson correlation (r)	2
Qualitative (3+ groups)	Quantitative	ANOVA + post-hoc	3
Several quantitative	Quantitative	Multiple regression	4

Read this table again. Choosing the right test is not a matter of taste or memory — it is **mechanically determined** by the nature of your variables.

This is why the dataset you find today must contain the right *types* of variables.



Describing one variable

What “univariate” means

Univariate analysis = describing **one variable at a time**.

No relationships, no causes, no predictions — yet.

Qualitative variable	Quantitative variable	Both
Frequency table	Mean, median, mode	Count of valid cases
Percentages	Standard deviation	Missing values
Mode	Min–max, quartiles	
Bar chart / pie chart	Skewness, kurtosis	
	Histogram, boxplot	

The most common error in this course: computing a **mean** for a qualitative variable.

“Average gender = 1.79” is meaningless.

Frequency tables and the mode

Example: brand purchased ($N = 325$)

Category	Frequency	%
National brand	142	43.7
Regional brand "Grain d'Or"	183	56.3
Total	325	100

Mode

The **most frequent** category or value.

Here: *Regional brand*, with 183 observations (56.3%).

The mode is the only measure of central tendency valid for nominal data.

You cannot compute a mean or a median of "sector" — but you can say which sector is most common.

Central tendency: three different answers

Mean

Arithmetic average.

- + Uses all the data
- + Mathematically tractable
- Destroyed by outliers
- Misleading if skewed

Median

Middle value.

- + **Robust to outliers**
- + Works on ordinal data
- Ignores magnitudes
- Less convenient in models

Mode

Most frequent value.

- + Only option for nominal
- + Identifies the typical case
- May not be unique
- Ignores the rest

Exam question: “What are the main advantages and disadvantages of the mean, and which statistic avoids these disadvantages?”

Answer: the **median** (robust to outliers and skew). Confirm the shape with a **normality test** — Shapiro–Wilk or Kolmogorov–Smirnov.

Dispersion: the number managers forget

Two shops. Both average **50 €** per basket.

	Shop A	Shop B
Mean	50 €	50 €
Std. deviation	4 €	35 €

The **same mean**. Completely different businesses.

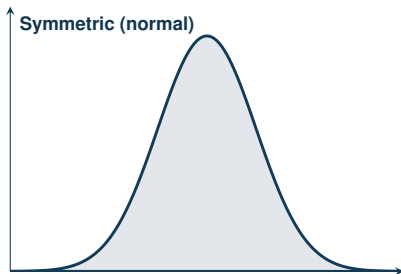
Standard deviation = the typical distance of an observation from the mean.

Small SD → homogeneous customers. One offer works for everyone.

Large SD → heterogeneous market. You must **segment** or you will satisfy nobody.

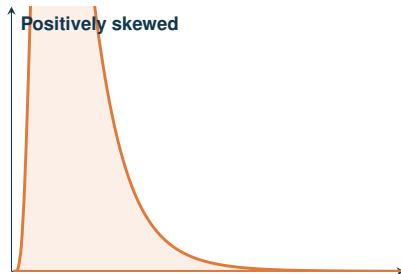
The mean tells you *where* the market is. The standard deviation tells you **whether the mean means anything at all**.

The shape of a distribution



Symmetric

mean $\approx$ median $\approx$ mode
skewness ≈ 0



Skewed right (typical of income, turnover)

mean $>$ median $>$ mode
skewness > 0 — a few large values pull the mean up

Skewness = asymmetry. **Kurtosis** = how heavy the tails are.
Rule of thumb: both between -1 and $+1$ is consistent with normality.

Is my variable normally distributed?

Three converging pieces of evidence — use **all three**, never just one:

1. **Compare the three centres.**

If $\text{mean} \approx \text{median} \approx \text{mode}$, the distribution is roughly symmetric. Large divergence signals skew.

2. **Check skewness and kurtosis.**

Both roughly within ± 1 .

3. **Look at the histogram.**

Does it actually look bell-shaped? Your eyes are a legitimate instrument.

Formal test: Shapiro–Wilk. Here H_0 = “the data *are* normal”.

So $p > 0.05$ means you **keep** normality — the opposite logic to every other test in this course. Be careful.

Read output critically — a real example

Variable	N	Mean	Median	Mode	Std. dev.	Min–Max
Age	325	41.2	41	41.3	10.5	18–70
Monthly spending (€)	325	56.7	50	70	3.3	10–110

Two things are wrong here.

Can you spot them before I tell you?

Mode of age = 41.3 — age is recorded in whole years. A mode cannot be fractional.

SD of spending = 3.3 — with a range of 10–110 €? Impossible.

Software never refuses to print a number. Sanity-checking output is your job.

The limit of univariate analysis

Knowing that average spending is **56.70 €** tells you **nothing** about *who* spends it, or *why*.

Univariate analysis describes variables **in isolation**. It cannot:

- establish a **relationship** between two variables
- compare **groups**
- identify what **drives** an outcome
- support any **prediction**

That is why we need sessions 2, 3 and 4.

And it is the answer to the exam question “why are univariate analyses insufficient?”



Finding your own data

The “fil rouge” project

You find one real dataset today. You analyse it for four weeks. It becomes a consulting report — and your coursework mark.

Artefact	Content	Marks
1 (today)	Clean database + univariate description	20
2	Chi-square + correlation	25
3	ANOVA + post-hoc	20
4	Regression + consolidated recommendations	25
Provenance	Source quality, assessed today	10

Choose a topic you can **stand to look at for four weeks**.

Sustainability, CSR, green business and retail are especially welcome — they are part of the course design.

Dataset specification — all boxes must be ticked

- ☐ **≥ 100 observations** (rows)
- ☐ **≥ 1 nominal variable with 2–5 categories**
your grouping variable for sessions 2 and 3 — without it those labs are impossible
- ☐ **≥ 1 ordinal variable**
- ☐ **≥ 3 continuous variables**
your correlation and regression material for sessions 2 and 4
- ☐ **One row = one** _____ — answerable in one sentence
- ☐ **A real downloadable file** (.csv or .xlsx) — not a screenshot, not a chart
- ☐ **Provenance:** URL, producer, year, sample size

Banned: Titanic, Iris, Boston Housing, anything labelled “cleaned” or “for beginners”.
Pre-processed data teaches you nothing.

Where to find financial data — the map



For this course, company-level data is the richest choice. One row = one firm (or one firm-year), with sector as your nominal variable and turnover, ROI, headcount and emissions as your continuous variables.

That structure supports **all four** sessions.

Company & financial data — where to actually go

Source	What you get	Access
Diane	French company accounts: turnover, net profit, headcount, ratios	EM Normandie library
Orbis (BvD)	Same, worldwide, 400M+ companies	EM Normandie library
Infogreffe / Pappers	French company filings, free tier	<code>pappers.fr</code>
SEC EDGAR	US-listed filings (10-K, financials), free	<code>sec.gov/edgar</code>
Yahoo Finance	Prices, key ratios, some ESG; easy CSV export	<code>finance.yahoo.com</code>
Euronext	Listed companies Paris/Amsterdam/Brussels	<code>live.euronext.com</code>
LSEG / Bloomberg	Professional terminals, everything	If your campus has a terminal

Start with Diane or Orbis. Ask the Learning Centre for access. You can export a sector, a size band and a year directly to Excel — that is a ready-made dataset of several hundred firms.

Macro, sector and open data

Source	What you get	Access
INSEE	French demographics, household spending, firm demography	<code>insee.fr</code>
data.gouv.fr	French open data, all ministries — large and realistically messy	<code>data.gouv.fr</code>
Eurostat	EU-comparable statistics, excellent documentation	<code>ec.europa.eu/eurostat</code>
Banque de France	Webstat: rates, credit, business surveys	<code>webstat.banque-france.fr</code>
ECB Data Portal	Euro area monetary & financial statistics	<code>data.ecb.europa.eu</code>
World Bank	Country indicators, 1960–present	<code>data.worldbank.org</code>
OECD Data Explorer	Cross-country economic and social data	<code>data-explorer.oecd.org</code>
IMF	Balance of payments, fiscal, financial soundness	<code>data.imf.org</code>

Careful: macro data often gives you one row = one country-year. With 27 EU countries you have few rows. Combine several years, or go to firm level.

ESG, CSR and sustainability data

Source	What you get	Access
CDP	Corporate climate, water & forest disclosures	cdp.net
ADEME	French emissions factors, Bilan GES of organisations	bilans-ges.ademe.fr
Our World in Data	Emissions, energy, food systems; clean CSV export	ourworldindata.org
EU ETS registry	Verified emissions per installation, per year	climate.ec.europa.eu
Yahoo Finance	Sustainability tab: ESG risk scores by firm	finance.yahoo.com
CSRD reports	Since 2024, large EU firms publish standardised sustainability data	Company websites

A strong project shape: one row = one listed firm.

Nominal: *sector*. Ordinal: *ESG grade*. Continuous: *turnover*, *ROI*, *emissions per employee*.

Question: **do greener firms perform better financially?** That carries all four sessions.

General repositories — and how to judge a source

Repositories

- **Google Dataset Search**
`datasetsearch.research.google.com`
- **Kaggle Datasets**
but avoid the “beginner” sets
- **UCI ML Repository**
- **Harvard Dataverse**
replication data from published research
- **Zenodo**

Before you download, ask:

1. **Who** produced it? Statistical office, regulator, company, anonymous user?
2. **When** was it collected? Not published — *collected*.
3. **How** was it collected? Survey, administrative record, scraping?
4. **Is there a codebook** explaining what each column means?
5. **What is missing**, and why?

“Garbage in, garbage out.” No statistical method can rescue a bad dataset.
Ten minutes spent judging the source saves four weeks of wasted work.

Approval slip — complete before you leave

Team members

Dataset title

Source URL

Producer & year

One row = one...

N (rows)

Nominal variable (2–5 categories)

Ordinal variable

Continuous variables (≥ 3)

Managerial question

Lecturer signature

No signature $\Rightarrow$ you receive the backup dataset and forfeit the 10 provenance marks.



Jamovi lab

Why Jamovi

- Free and open source — jamovi.org
- Installs in five minutes, Mac & Windows
- Output tables look like SPSS, so everything transfers
- Results update **live** as you change the data

Two tabs matter

- **Data** — your spreadsheet and variable settings
- **Analyses** — every test in this course

Import your file

≡ menu → **O**pen → **B**rowse → your .csv or .xlsx

Save your work

≡ menu → **S**ave **A**s → .omv

The .omv file stores data *and* analyses together.
That is what you submit.

Setting up your variables — do not skip this

Double-click any column header in the **Data** tab:

1. **Name** — short, no spaces, no accents
2. **Description** — what it actually measures, and in which units
3. **Measure type** — **the critical setting**:
 - Nominal** — categories, no order
 - Ordinal** — ordered categories
 - Continuous** — numeric measurements
 - ID** — identifiers, never analysed
4. **Levels** — rename and reorder categories here

If the measure type is wrong, Jamovi will silently **refuse** to offer you the right test in week 2.
This is the number one cause of a lost lab session.

Cleaning: what to hunt for

- **Missing values in disguise**
99, -1, 9999, "N/A", empty strings
- **Duplicate rows**
the same firm listed twice
- **Impossible values**
age 150, negative turnover, ROI of 4000%
- **Inconsistent categories**
"Paris" / "paris" / "PARIS" / "Paris "
- **Wrong decimal separator**
French data uses commas: 56,7 not 56.7
- **Numbers stored as text**
Jamovi will treat them as nominal

Keep a cleaning log. What you found, what you did, why.
This is graded — and in professional practice it is what makes your analysis defensible.

Running your univariate analysis

Analyses → Exploration → Descriptives

For qualitative variables

- Tick Frequency tables
- Read counts and percentages
- Identify the **mode**
- Plots: Bar plot

For quantitative variables

- Statistics: mean, median, mode, SD, min, max
- Tick **skewness** and **kurtosis**
- Plots: Histogram + Box plot
- Split by a grouping variable if useful

Then write it up in the **protocol** format. Not “the mean is 56.7”, but
“spending is right-skewed, so the median of 50 € better represents the typical customer; the high SD suggests two distinct segments.”

Artefact 1 — submit before you leave (20 marks)

1. The cleaned **.omv** file
2. The completed **approval slip** (provenance)
3. A **cleaning log**: what was wrong, what you did about it
4. **Univariate description of five variables**, written out in sentences
5. For one continuous variable: **is it normal?** Justify with the three pieces of evidence
6. **Two sentences** on what a manager learns from your descriptive statistics

Half the marks are for points **4–6** — the interpretation.
Producing output is not analysis. **Explaining it is.**

Errors that cost marks every year

- Reporting a **mean for a nominal variable** — “average sector = 2.4”
- Setting the **measure type** wrong, then wondering why the test will not run
- Reading “**Sig. = 0.000**” as $p = 0$ — it means $p < 0.001$
- Confusing **significance** (is there an effect?) with **strength** (does it matter?)
- Copying output into a report with **no interpretation**
- Stopping at the statistic. **The manager asked a question. Answer it.**

Before session 2

1. **Finish cleaning** your dataset. Session 2 assumes it is ready.
2. **Identify two pairs of variables** in your own data:
 - one pair of **qualitative** variables → we will run a chi-square
 - one pair of **quantitative** variables → we will run a correlation
3. **Write the managerial question** each pair could answer.
4. **Read:** Navarro & Foxcroft, *Learning Statistics with Jamovi*, chapters on descriptives.

You will arrive next week with a live question and the data to answer it.
That is how this course is supposed to feel.

Questions?

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Session 2: relationships between two variables